Frequency Response Descriptions for LTI Systems

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Key Concepts

1) The frequency response of a system tells us how the system responds to complex sinusoidal input signals as a function of their frequency.

2) If the input is a complex sinusoid $x[n] = e^{j\omega n}$, the output is also a complex sinusoid $y[n] = H(e^{j\omega})e^{j\omega n}$ where $H(e^{j\omega})$ is a complex number representing the frequency response.

    a) The relationship between the input and output sinusoids is multiplication by the complex number $H(e^{j\omega})$.

b) The magnitude of the input sinusoid is modified by the magnitude of the frequency response $|H(e^{j\omega})|$.

c) The phase of the input sinusoid is shifted by the phase of the frequency response $\angle H(e^{j\omega})$.

3) If the input is a sum of complex sinusoids, the output is a sum of complex sinusoids modified by the corresponding frequency response of the system.

4) The frequency response is the discrete-time Fourier transform of the impulse response of the system.

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

5) The frequency response of a system is expressed in terms of the difference equation coefficients as

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

6) The frequency response describes the filtering properties of the system. That is, how it discriminates between or separates the different frequencies of the input signal.

    a) A lowpass filter passes low frequencies and attenuates high frequencies.

    b) A high-pass filter passes high frequencies and attenuates low frequencies.

c) A band-pass filter passes a range of frequencies while attenuating those above and below the pass range.

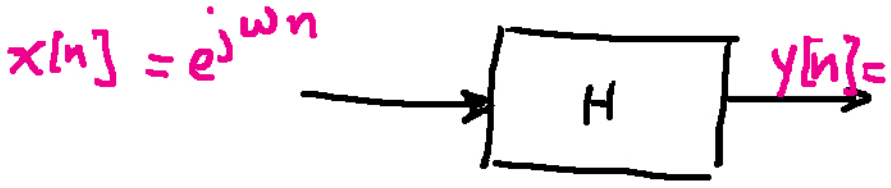
d) A band-stop filter passes low and high frequencies while attenuating those in a band.

e) Discrete-time filters are implemented by passing data through the difference equation or impulse response associated with the system.

f) Continuous-time filters are usually implemented with electrical circuits.

Video Screenshots

Frequency response: how system responds to sinusoids of different frequencies



Multiplication!

↖ freq response - independent of n

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j \measuredangle H(e^{j\omega})} \Rightarrow y[n] = |H(e^{j\omega})| e^{j \measuredangle H(e^{j\omega})} e^{j\omega n}$$

$$= |H(e^{j\omega})| e^{j(\omega n + \measuredangle H(e^{j\omega}))}$$

- Amplitude of $e^{j\omega n}$ changed by $|H(e^{j\omega})|$ - magnitude response
- Phase of $e^{j\omega n}$ changed by $\measuredangle H(e^{j\omega})$ — phase response

By linearity, if

$$x[n] = \alpha_1 e^{j\omega_1 n} + \alpha_2 e^{j\omega_2 n} \Rightarrow y[n] = \alpha_1 H(e^{j\omega_1}) e^{j\omega_1 n} + \alpha_2 H(e^{j\omega_2}) e^{j\omega_2 n}$$

Relationship to impulse response

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$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = e^{j\omega n}\underbrace{\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}}_{H(e^{j\omega})}$$

$$h[n] \xleftrightarrow{\text{DTFT}} H(e^{j\omega})$$

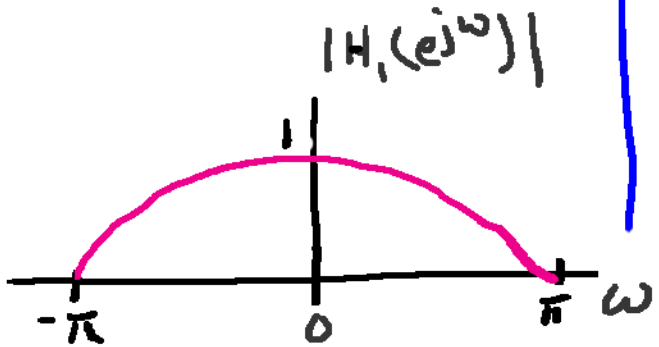
Examples:

$$y_1[n] = \tfrac{1}{2}x[n] + \tfrac{1}{2}x[n-1]$$

$$h_1[n] = \begin{cases} 1/2 & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

$$H_1(e^{j\omega}) = \tfrac{1}{2} + \tfrac{1}{2}e^{-j\omega}$$

$$= e^{-j\omega/2}\cos(\omega/2)$$

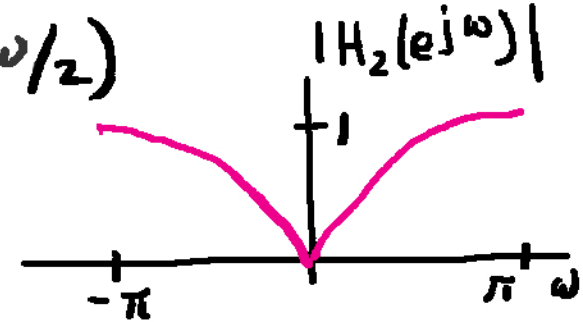


$$y_2[n] = \tfrac{1}{2}x[n] - \tfrac{1}{2}x[n-1]$$

$$h_2[n] = \begin{cases} 1/2 & n = 0 \\ -1/2 & n = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$H_2(e^{j\omega}) = \tfrac{1}{2} - \tfrac{1}{2}e^{-j\omega}$$

$$= je^{-j\omega/2}\sin(\omega/2)$$

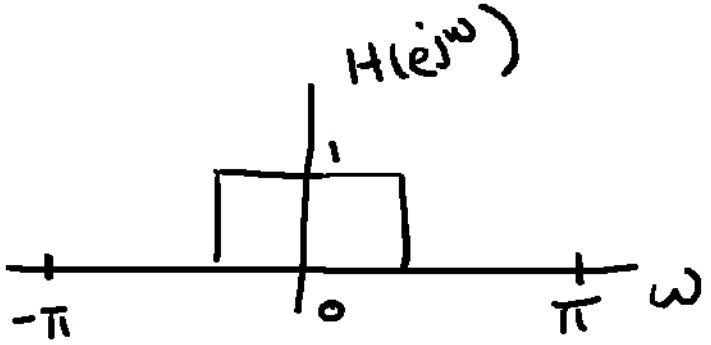


Convolution - Multiplication property

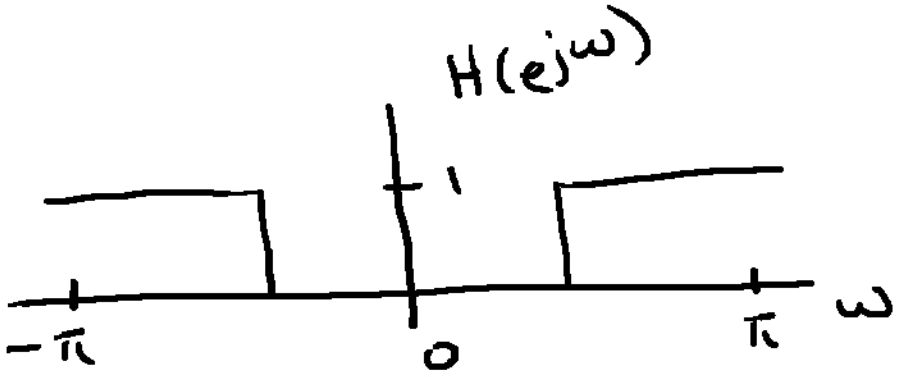
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$$y[n] = h[n] * x[n] \xleftrightarrow{\text{DTFT}} Y(e^{j\omega}) = H(e^{j\omega})\,X(e^{j\omega})$$

Ideal filters



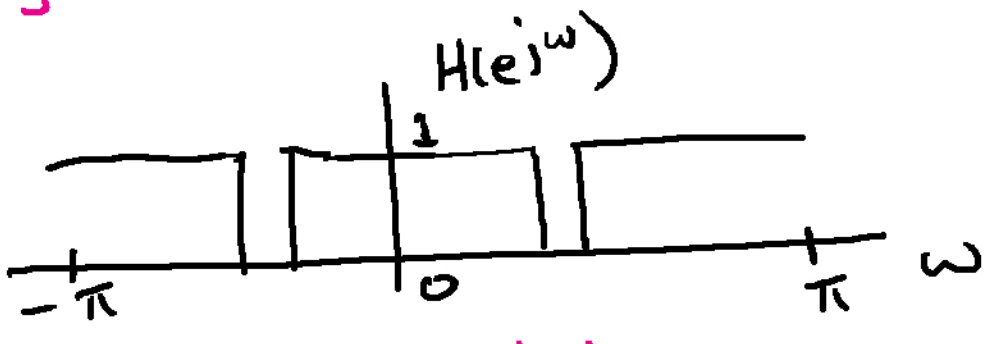
Lowpass



Highpass



Bandpass



Bandstop

Frequency response and difference equations

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$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \quad , \quad a_0 = 1$$

If $x[n] = e^{j\omega n}$, then $y[n] = H(e^{j\omega}) e^{j\omega n}$

$$\Rightarrow \sum_{k=0}^{N} a_k H(e^{j\omega}) e^{j\omega(n-k)} = \sum_{k=0}^{M} b_k e^{j\omega(n-k)}$$

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

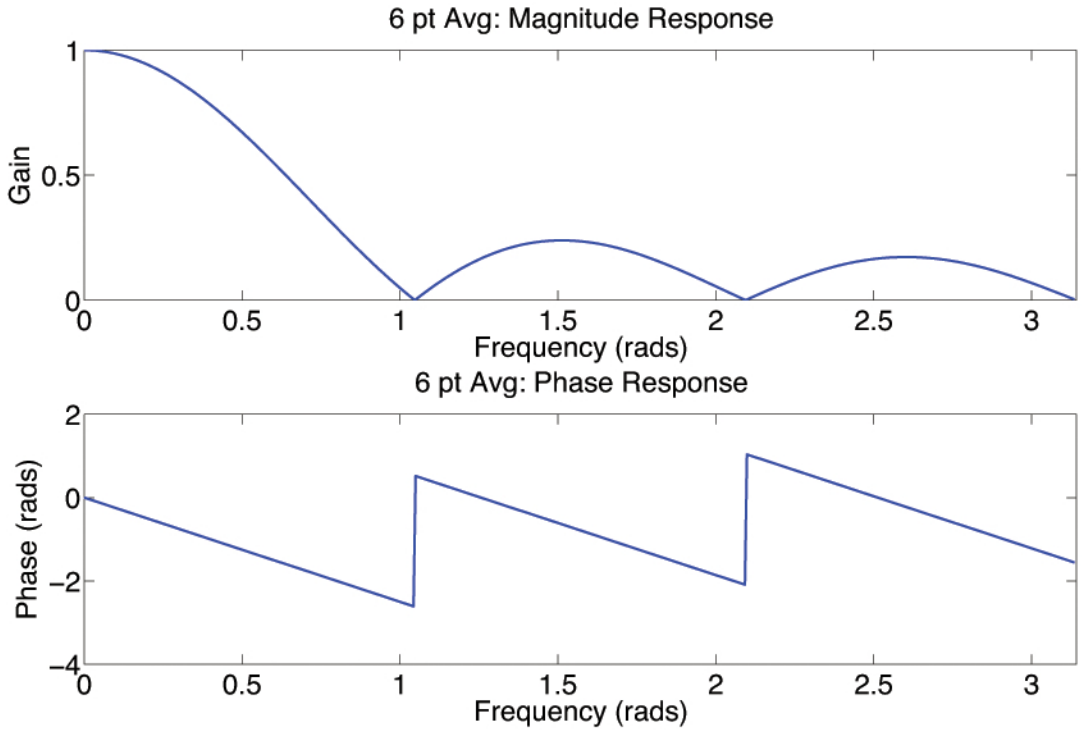
ratio of polynomials in $e^{-j\omega}$

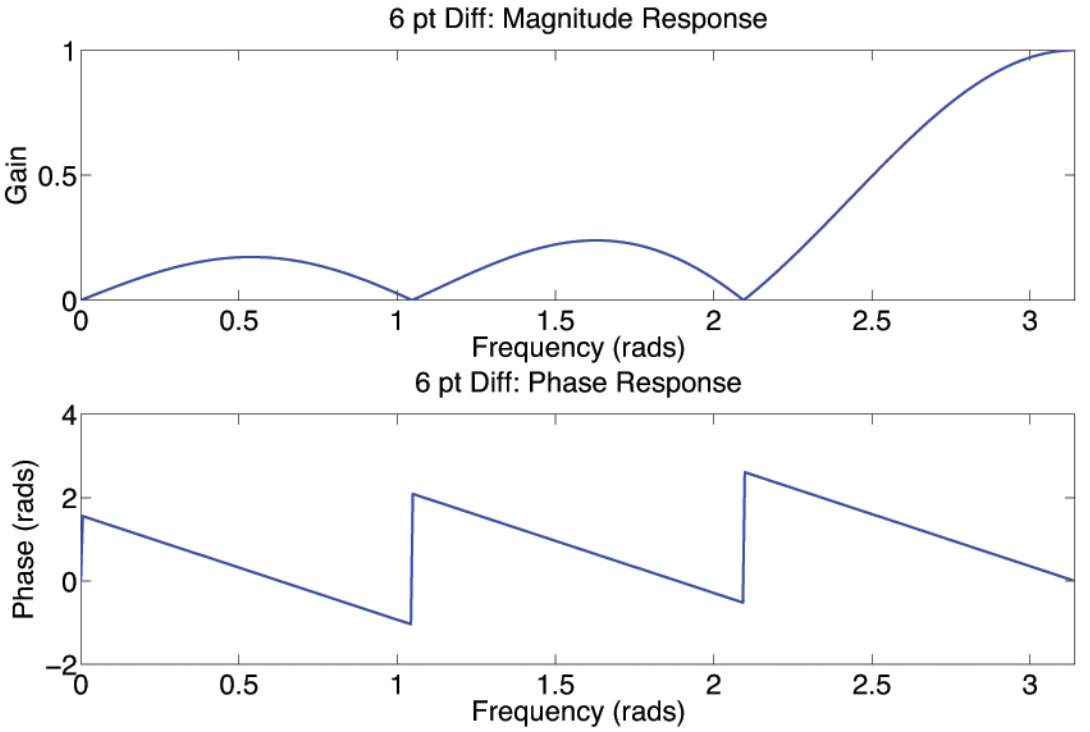
Examples:

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$$y[n] = \frac{1}{6}\{x[n] + x[n-1] + \ldots + x[n-5]\}$$

$$y[n] = \frac{1}{6}\{x[n] - x[n-1] + x[n-2] - x[n-3] + x[n-4] - x[n-5]\}$$



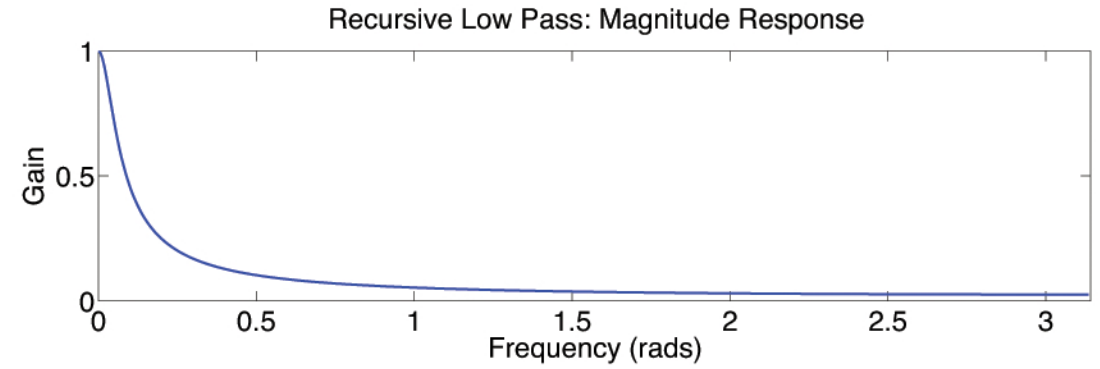


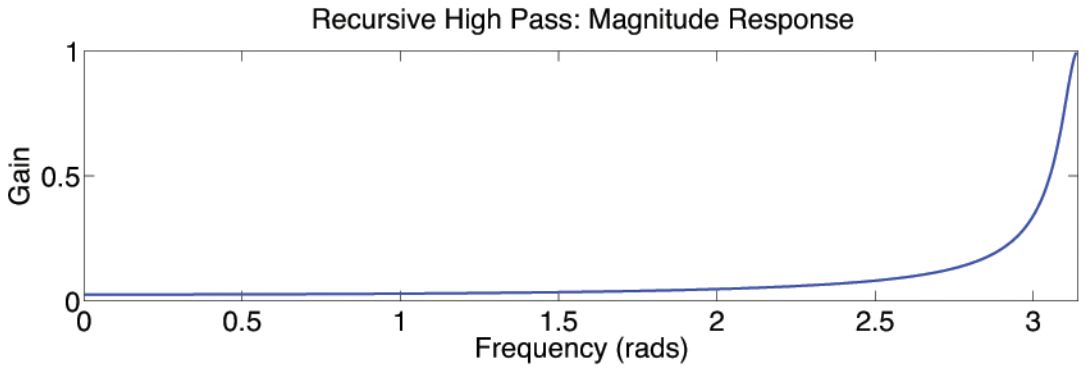
Examples:

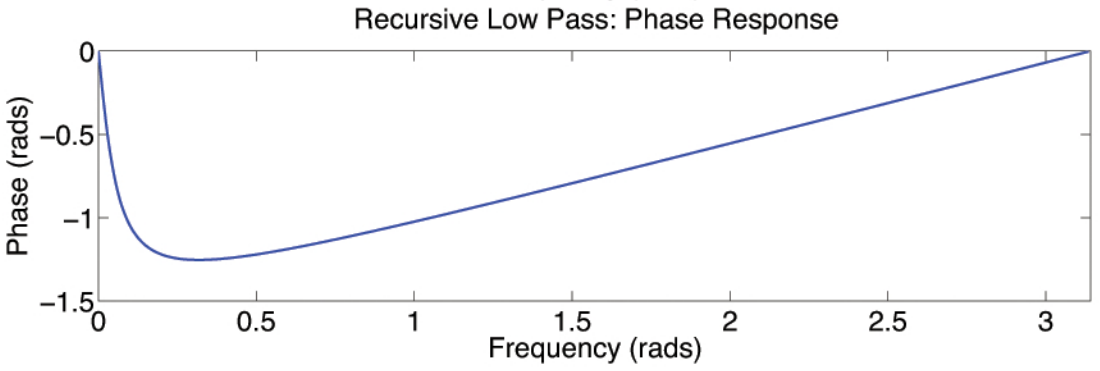
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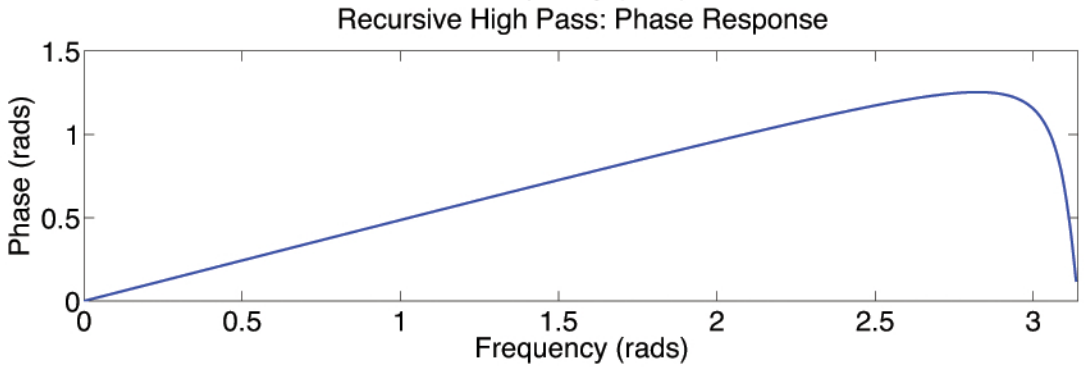
$$y[n] = 0.95\,y[n-1] + 0.05\,x[n]$$

$$y[n] = -0.95\,y[n-1] + 0.05\,x[n]$$

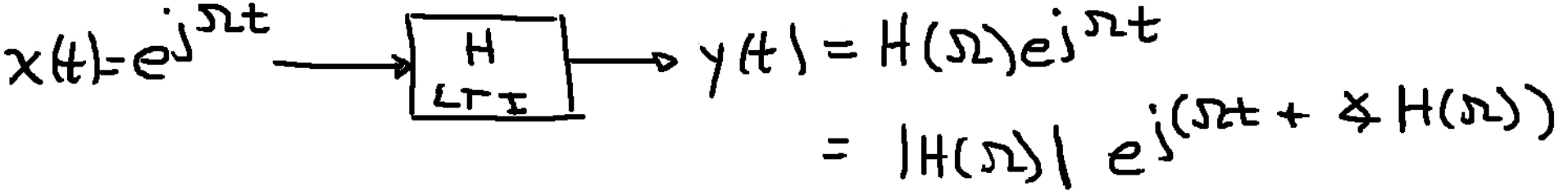








Frequency response and continuous-time systems



$$y(t) = h(t) * x(t) \quad \xleftrightarrow{FT} \quad Y(\Omega) = H(\Omega) X(\Omega)$$

$$h(t) \xleftrightarrow{FT} H(\Omega)$$

Filters - built using electrical circuits

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